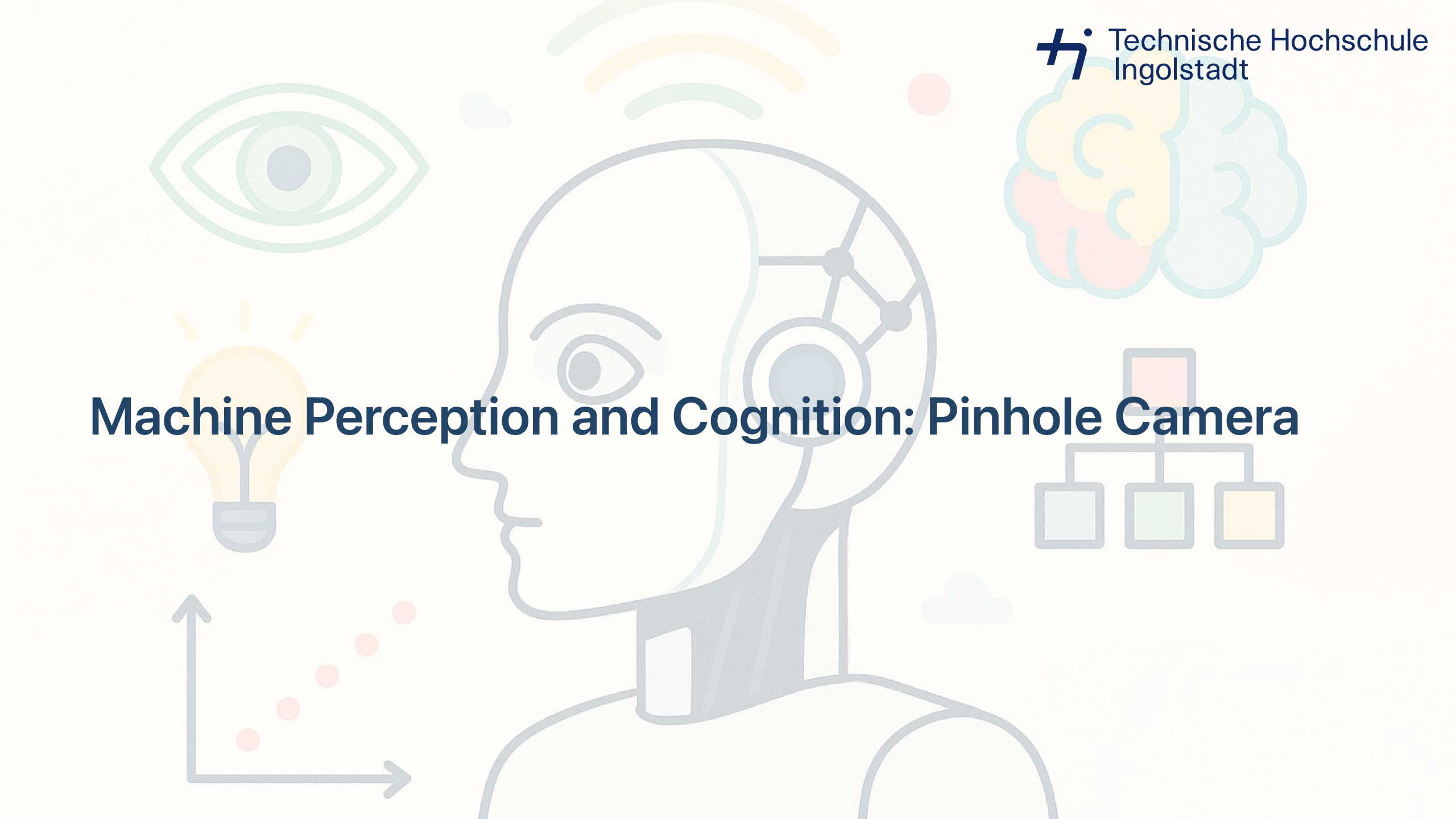
Technische Hochschule Ingolstadt



Machine Perception and Cognition: Pinhole Camera

Types of projections

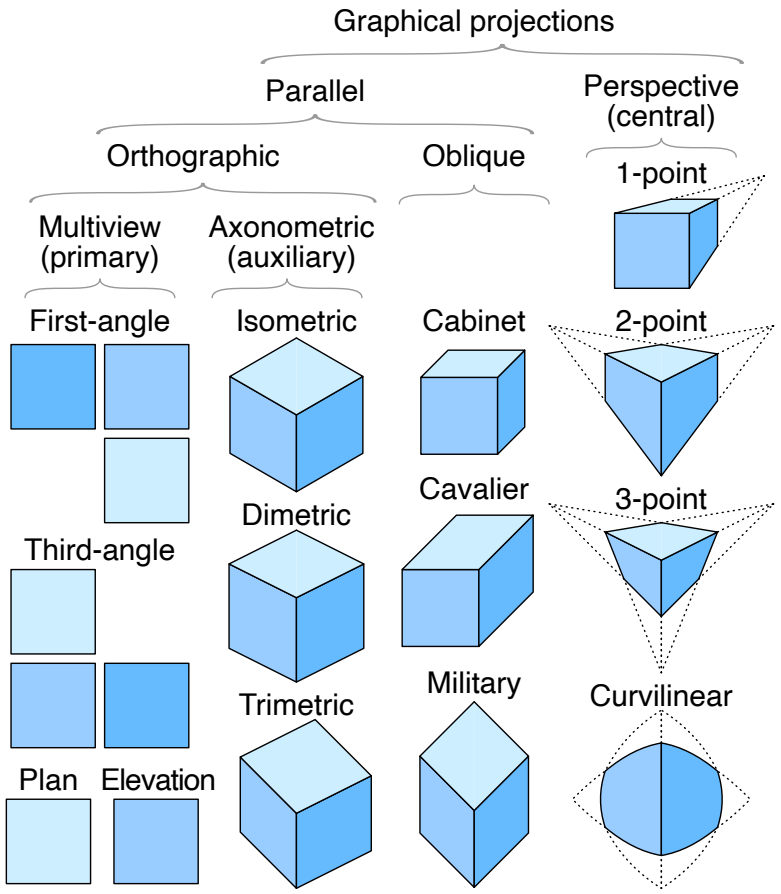


Image source: Cmglee, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Comparison_of_graphical_projections.svg

Perspective projection, here 2-point

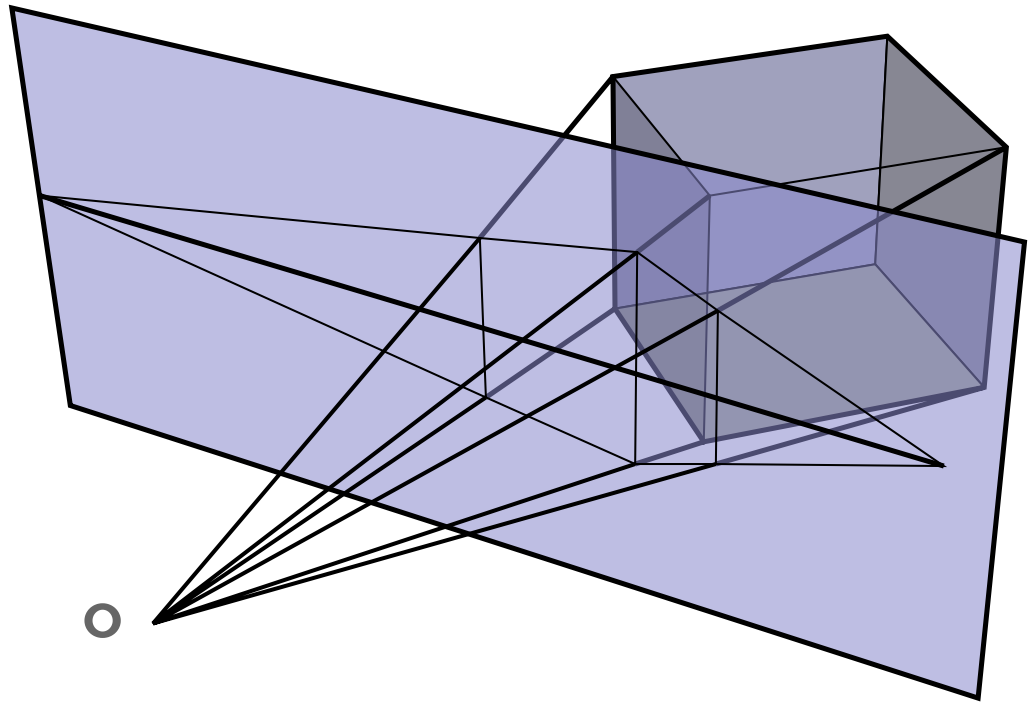


Image source: Marco Polo, CC BY 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://en.wikipedia.org/wiki/File:Perspectiva-2.svg



Disclaimer

The following derivation and formulations about the pinhole camera geometry and camera matrix are from the Wikipedia pages "Pinhole camera model" and "Camera matrix".

Image source: https://www.pexels.com/photo/brown-wooden-shelf-with-books-3646172/

Pinhole camera

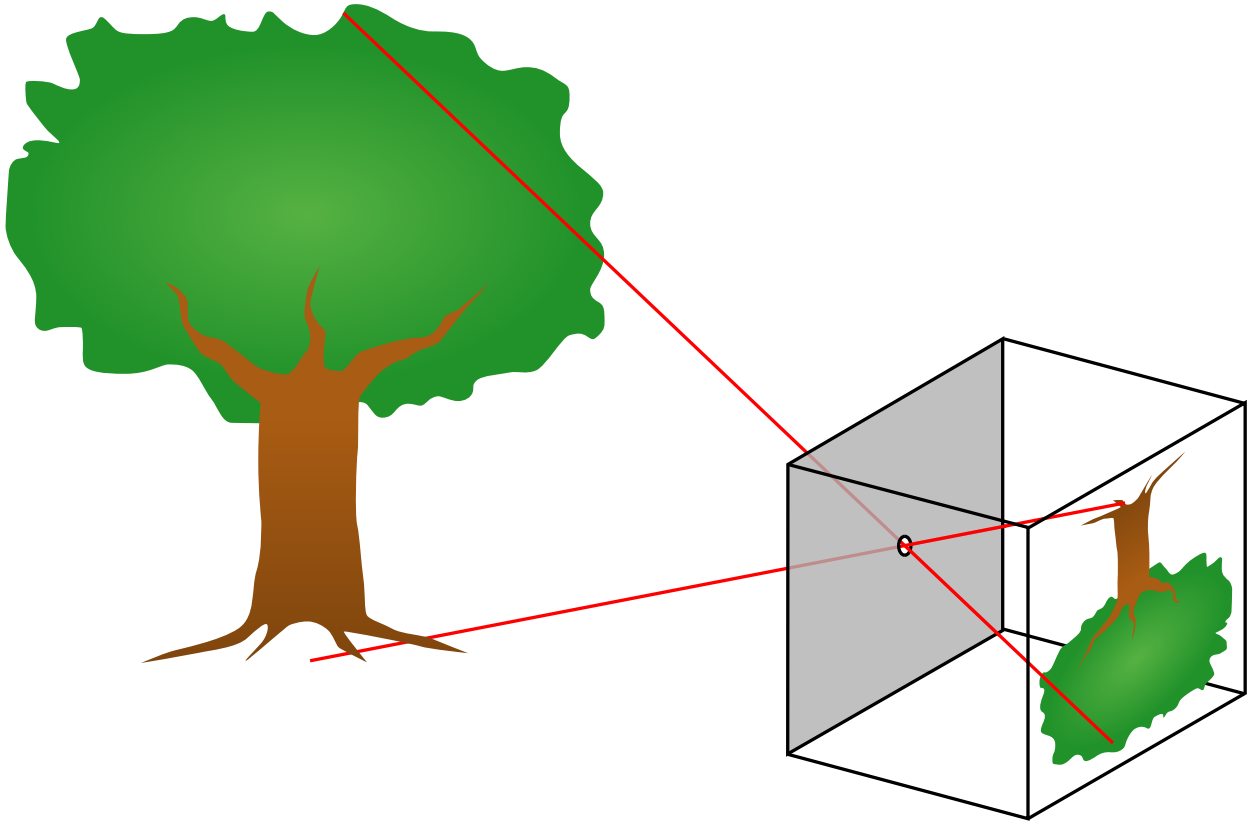


Image source: DrBob, Pbroks13, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole-camera.svg

Pinhole camera

From Wikipedia, page "Pinhole camera model":

> The **pinhole camera model** describes the mathematical relationship between the coordinates of a point in three-dimensional space and its projection onto the image plane of an ideal pinhole camera, where the camera aperture is described as a point and no lenses are used to focus light.

Pinhole camera

Geometric model of a perfect pinhole camera:

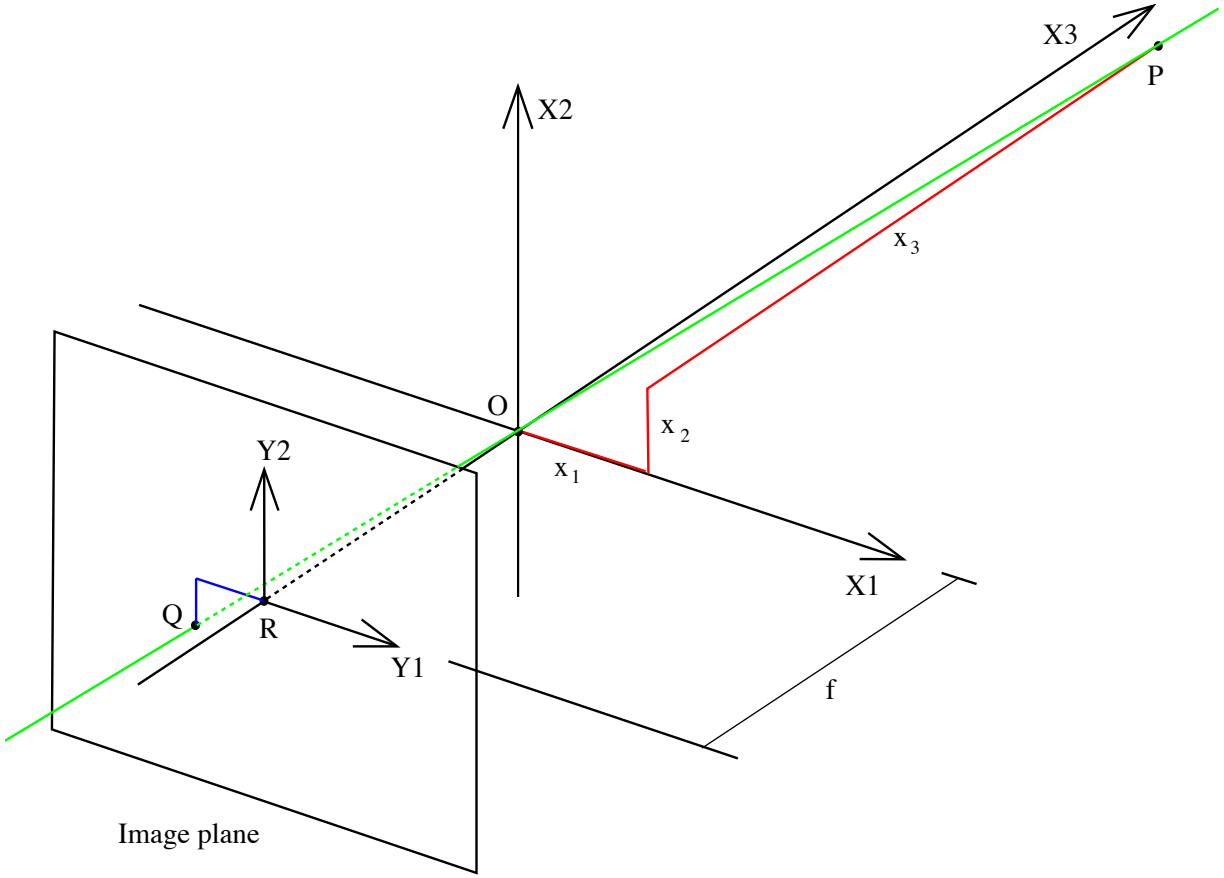


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

Axes X1, X2, X3 define the 3D *world coordinate system*:

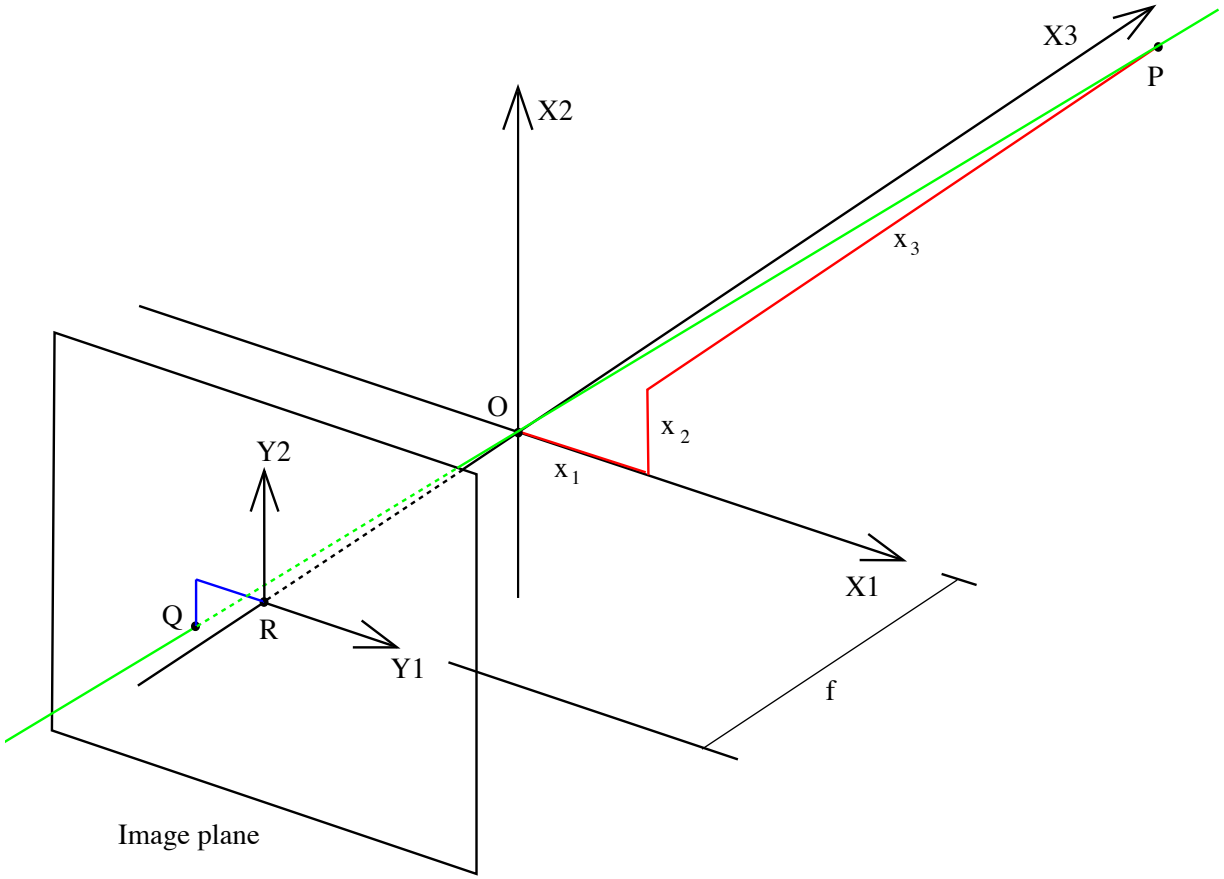


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

Axes Y1, Y2 define the 2D *image coordinate system*:

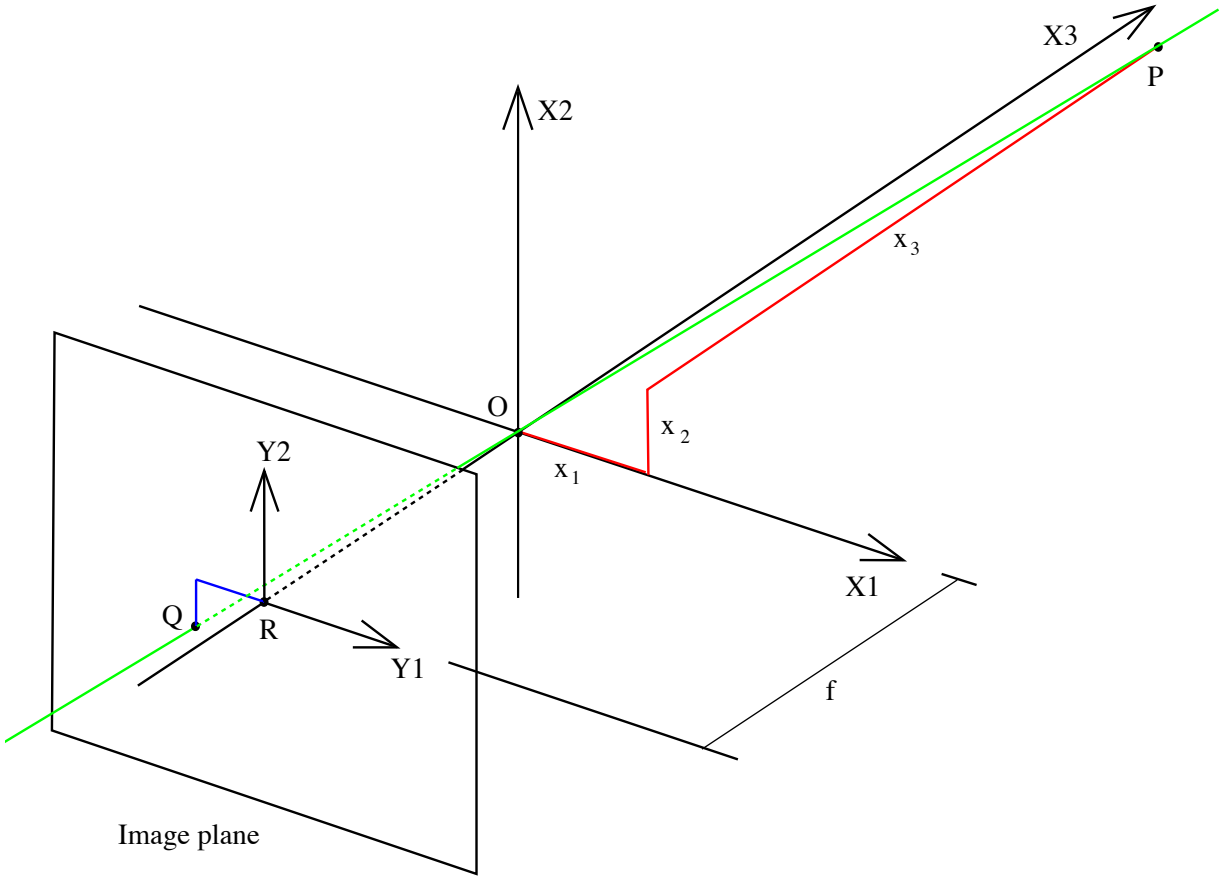


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

Origin O of the world coordinate system is also the *aperture* (pinhole) of the camera:

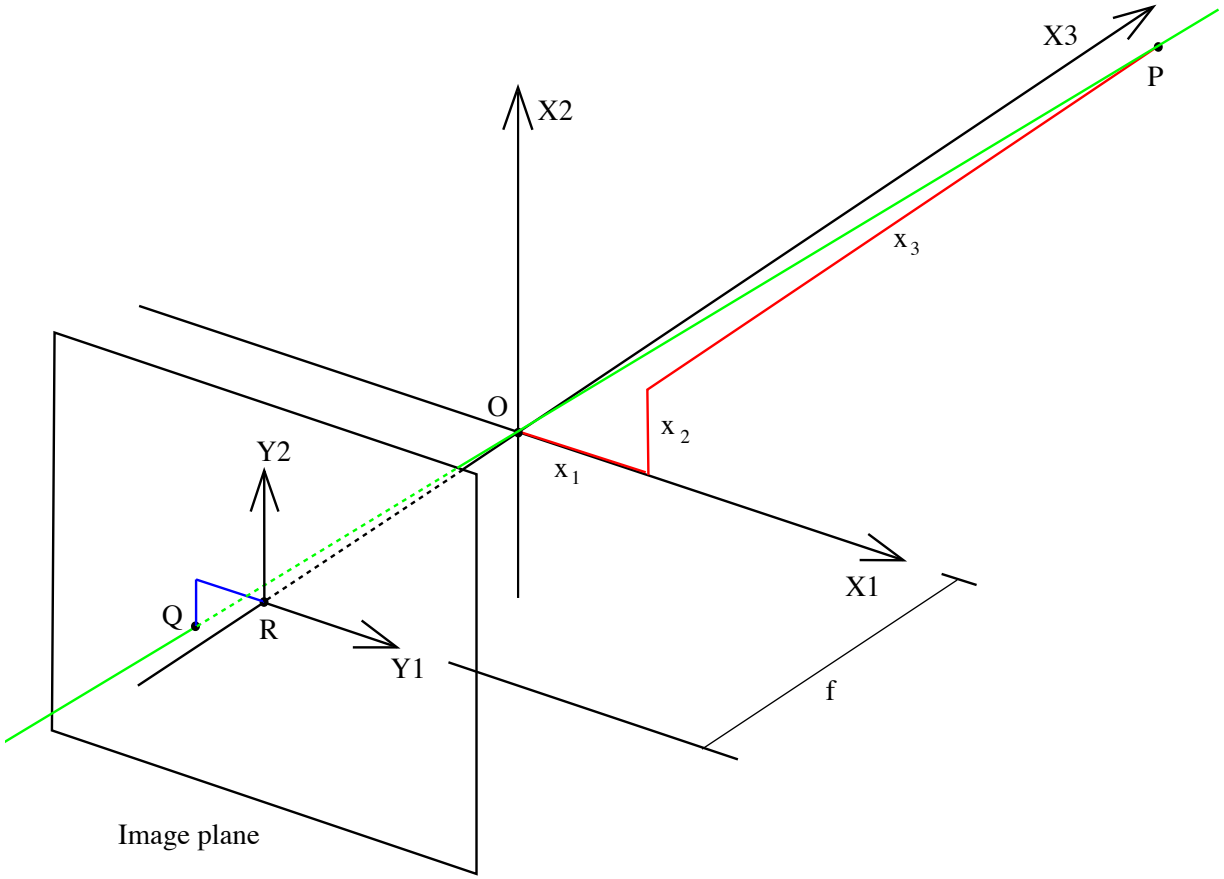


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

The *aperture* (point O) is in this model a point through which all projection lines must pass:

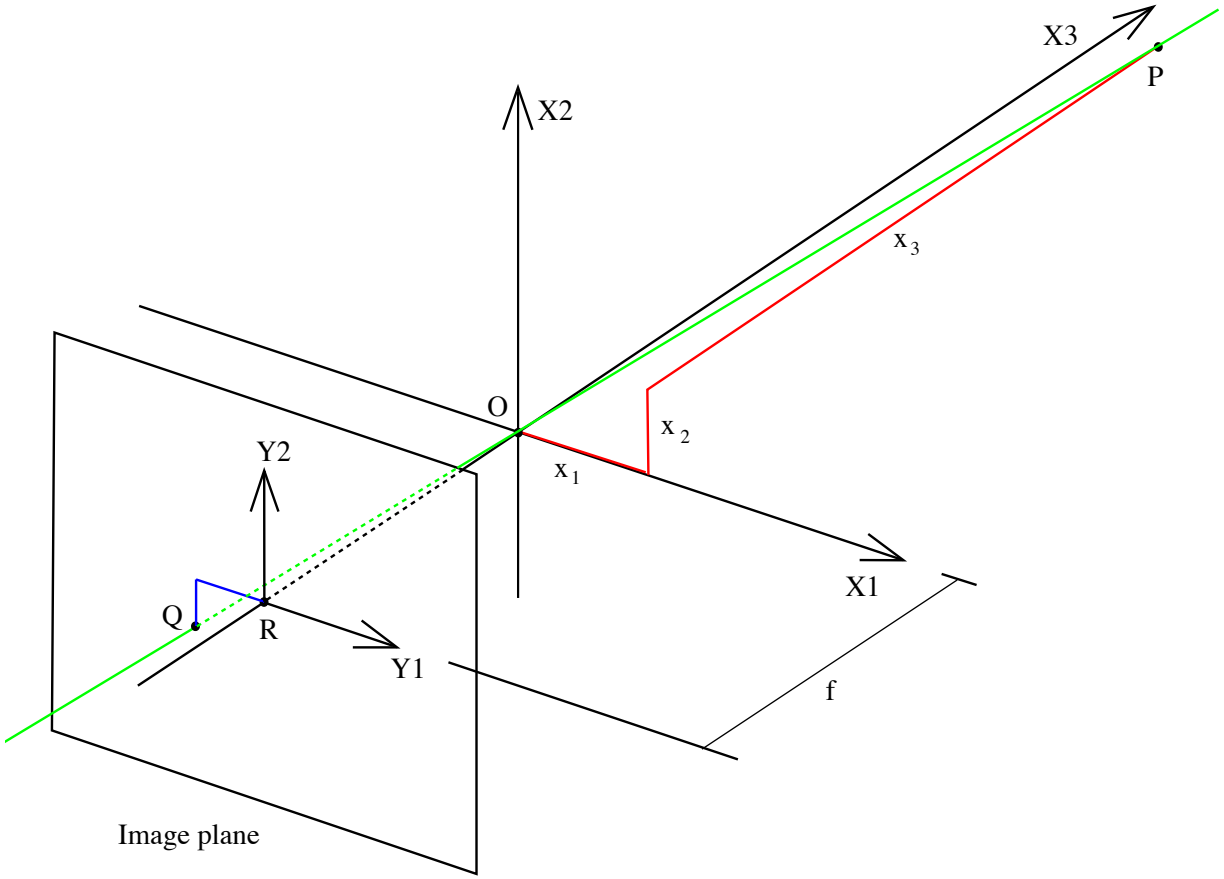


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

The *focal length* (distance f) is the distance between the aperture and the image plane:

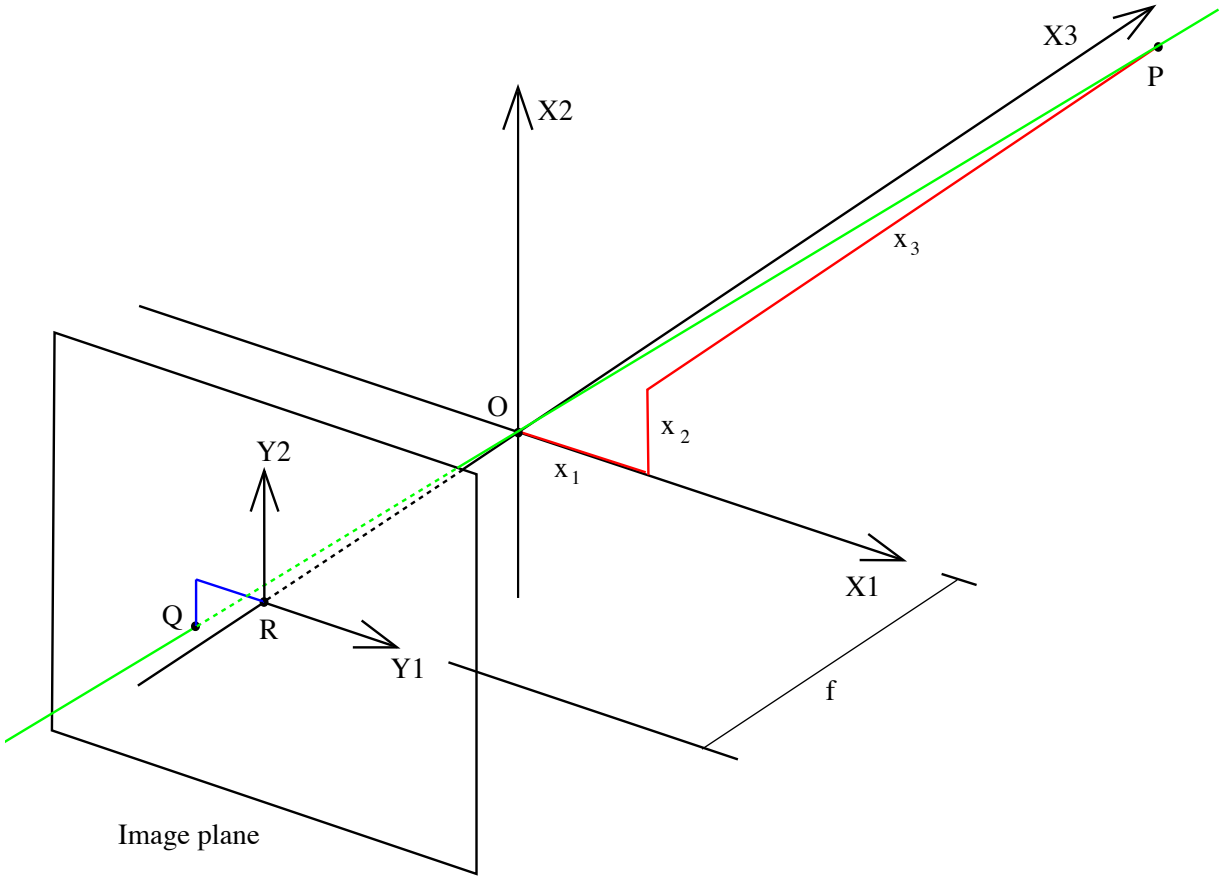


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

Point R is the intersection of the *optical axis* and the image plane:

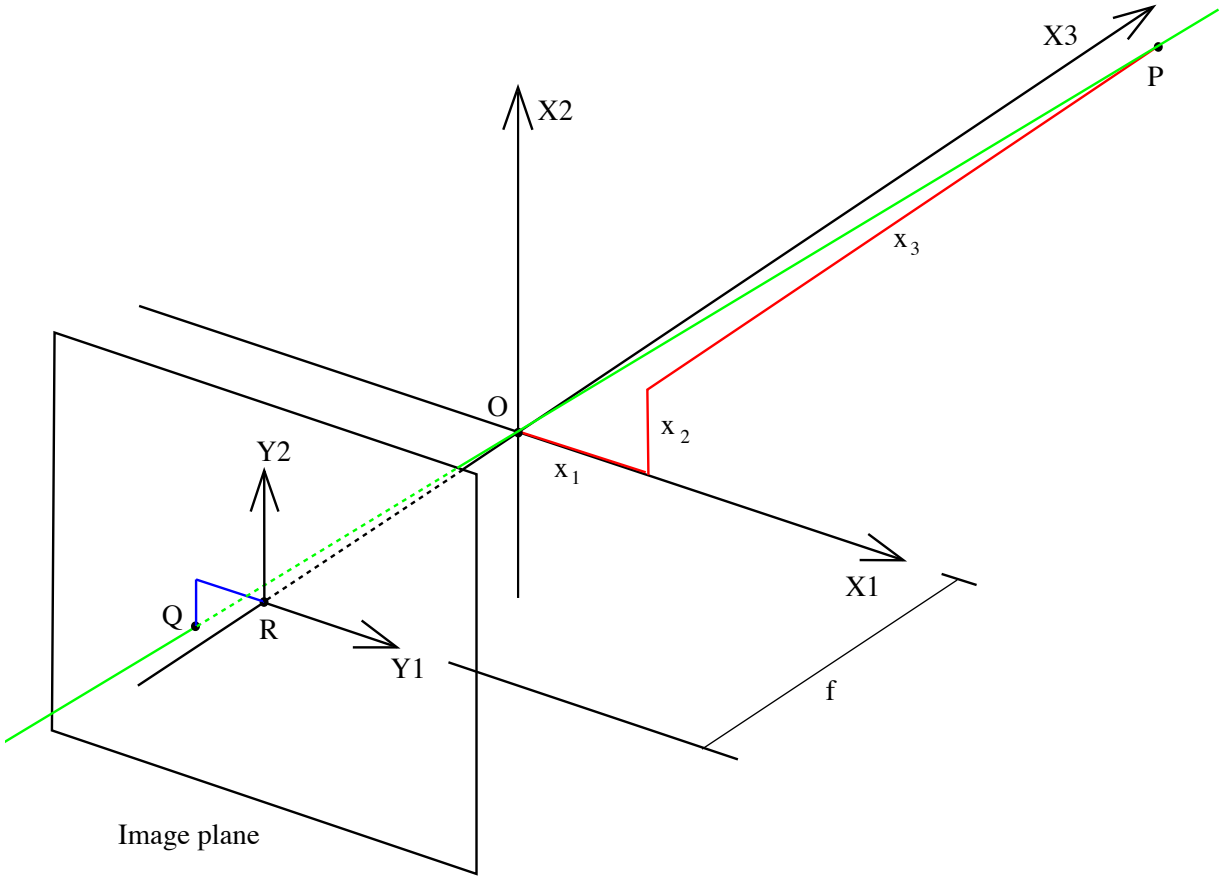


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

Point $P = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^\top$ is the 3D world point that is projected:

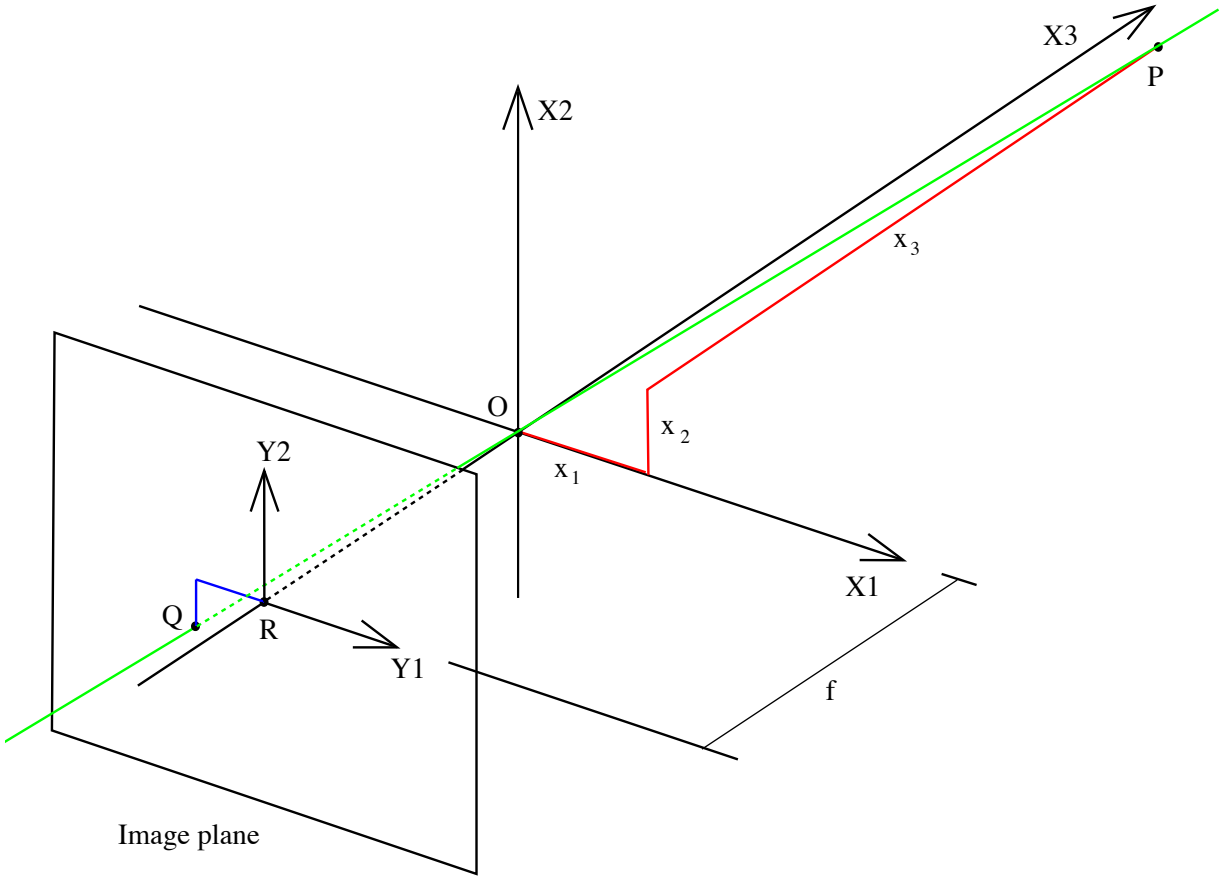


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

Point $Q = \begin{bmatrix} y_1 & y_2 \end{bmatrix}^\top$ is the projection of P onto the 2D image plane:

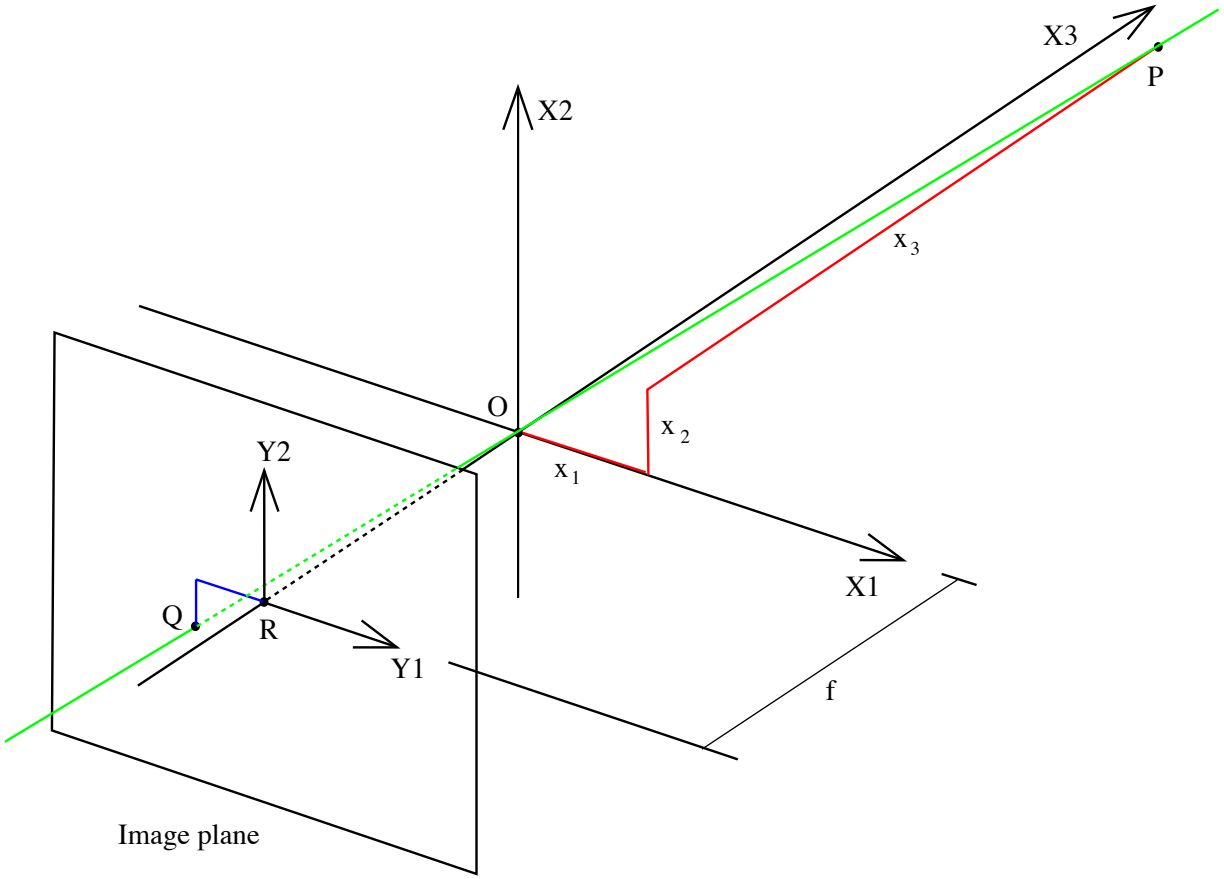


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

Please observe that in a pinhole camera, the projected image is mirrored:

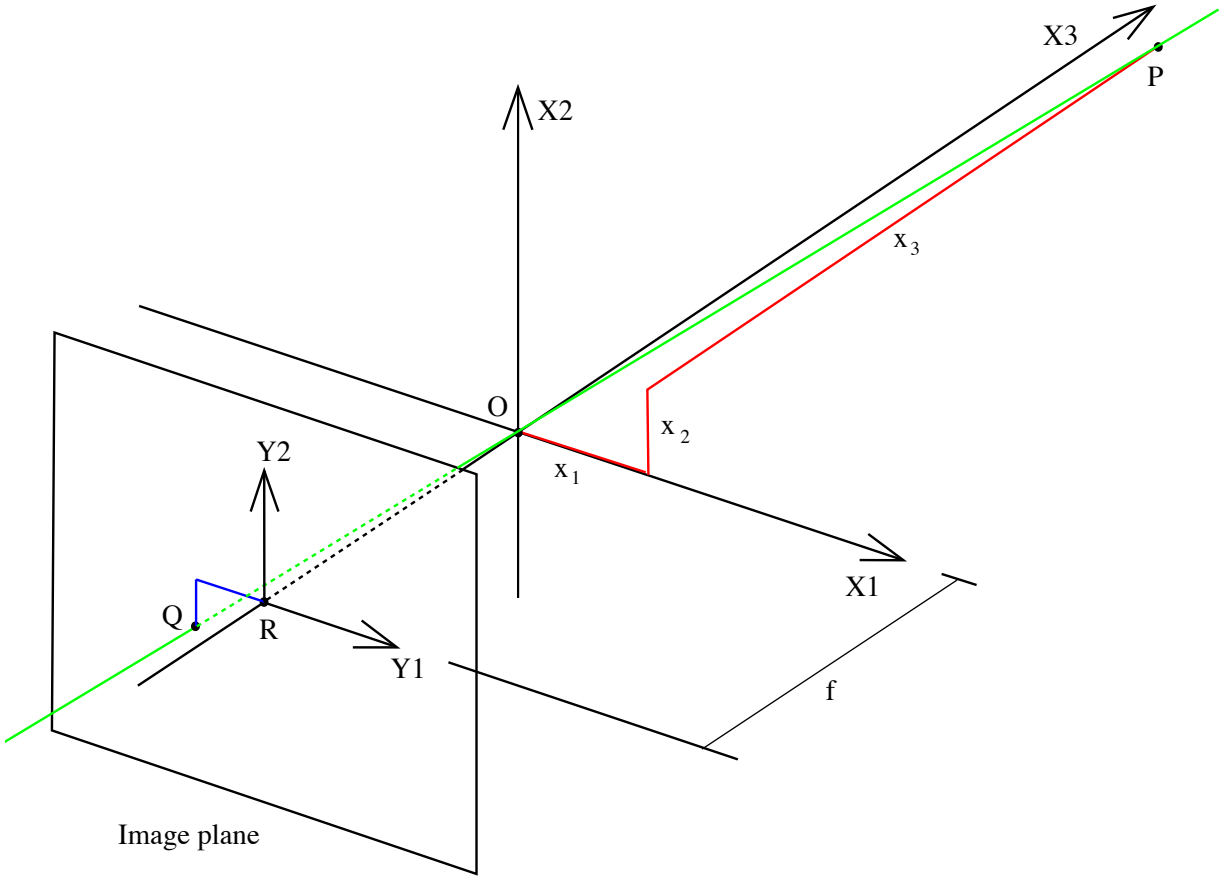


Image source: en:User:KYN, en:User:N3bulous, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole.svg

Pinhole camera

The same model as viewed along the X2 axis:

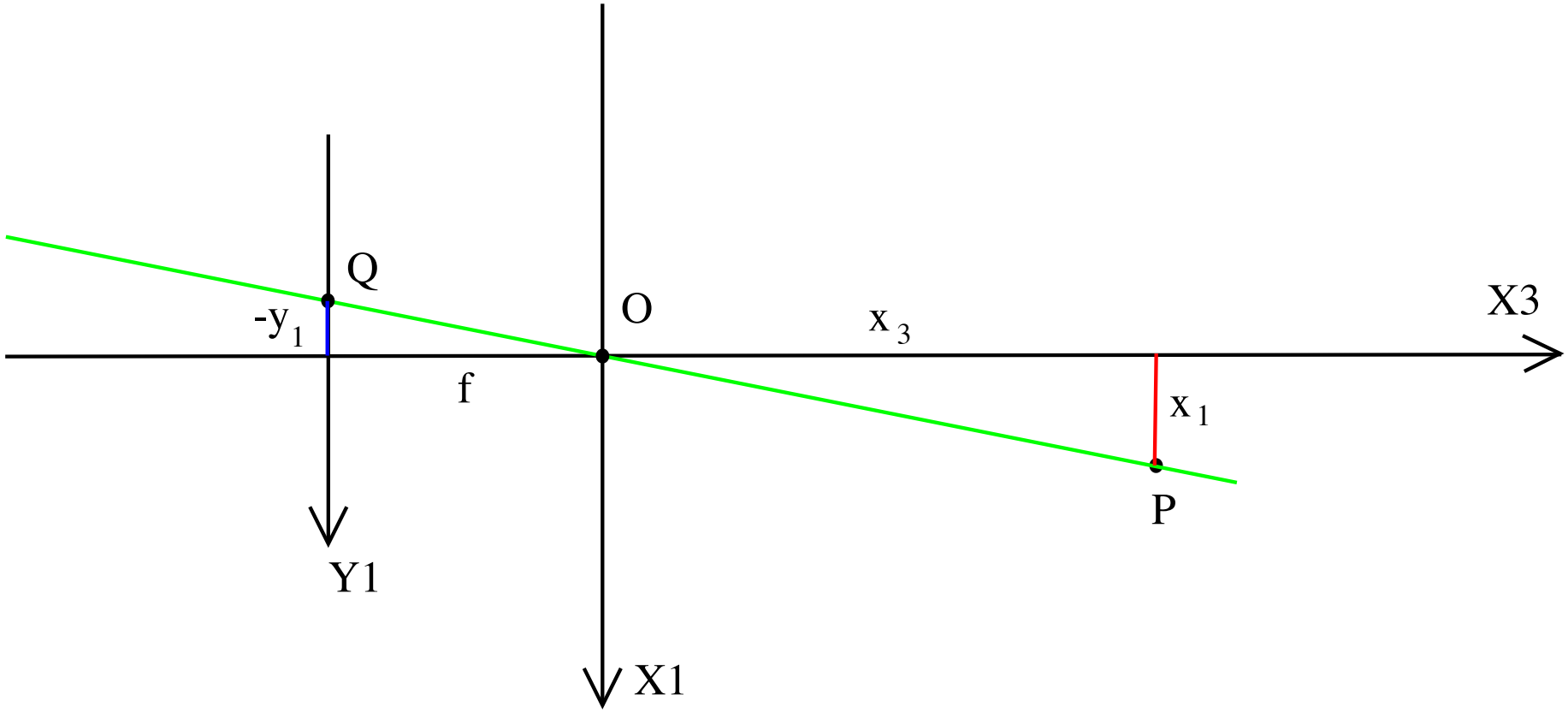


Image source: en:User:KYN, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole2.svg

Pinhole camera

The two triangles have identical angles between the optical axis and projection line:

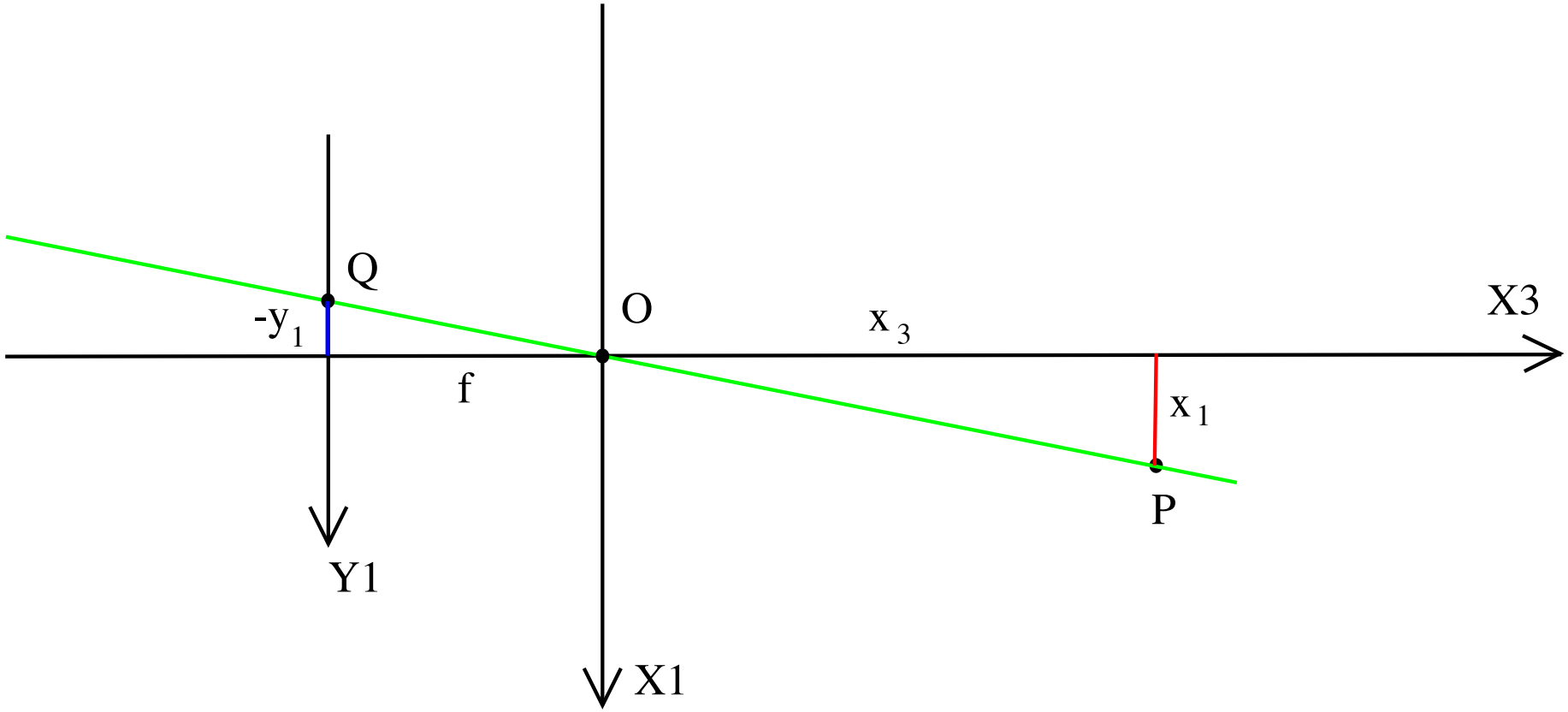


Image source: en:User:KYN, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole2.svg

Pinhole camera

Thus $\frac{-y_1}{f} = \frac{x_1}{x_3}$ or $y_1 = -\frac{f \times x_1}{x_3}$ assuming that $f > 0$ and $x_3 > 0$

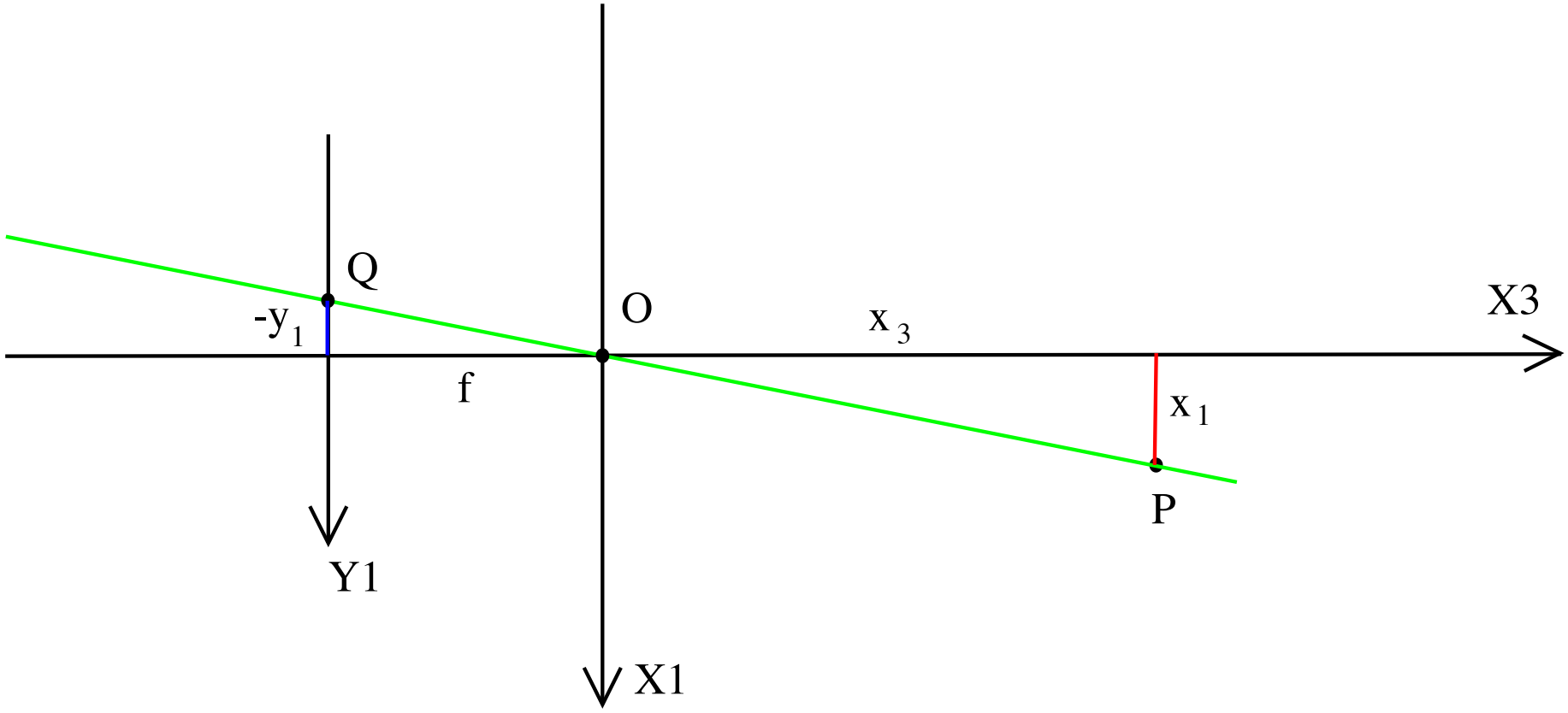


Image source: en:User:KYN, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole2.svg

Pinhole camera

The same relationship is true when viewed along the X1 axis:

$\frac{-y_2}{f} = \frac{x_2}{x_3}$ or $y_2 = -\frac{f \times x_2}{x_3}$

Pinhole camera

This relationship can be expressed for both dimension of the image plane:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = -\frac{f}{x_3} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Pinhole camera

Can we undo the mirroring over the image plane axes?

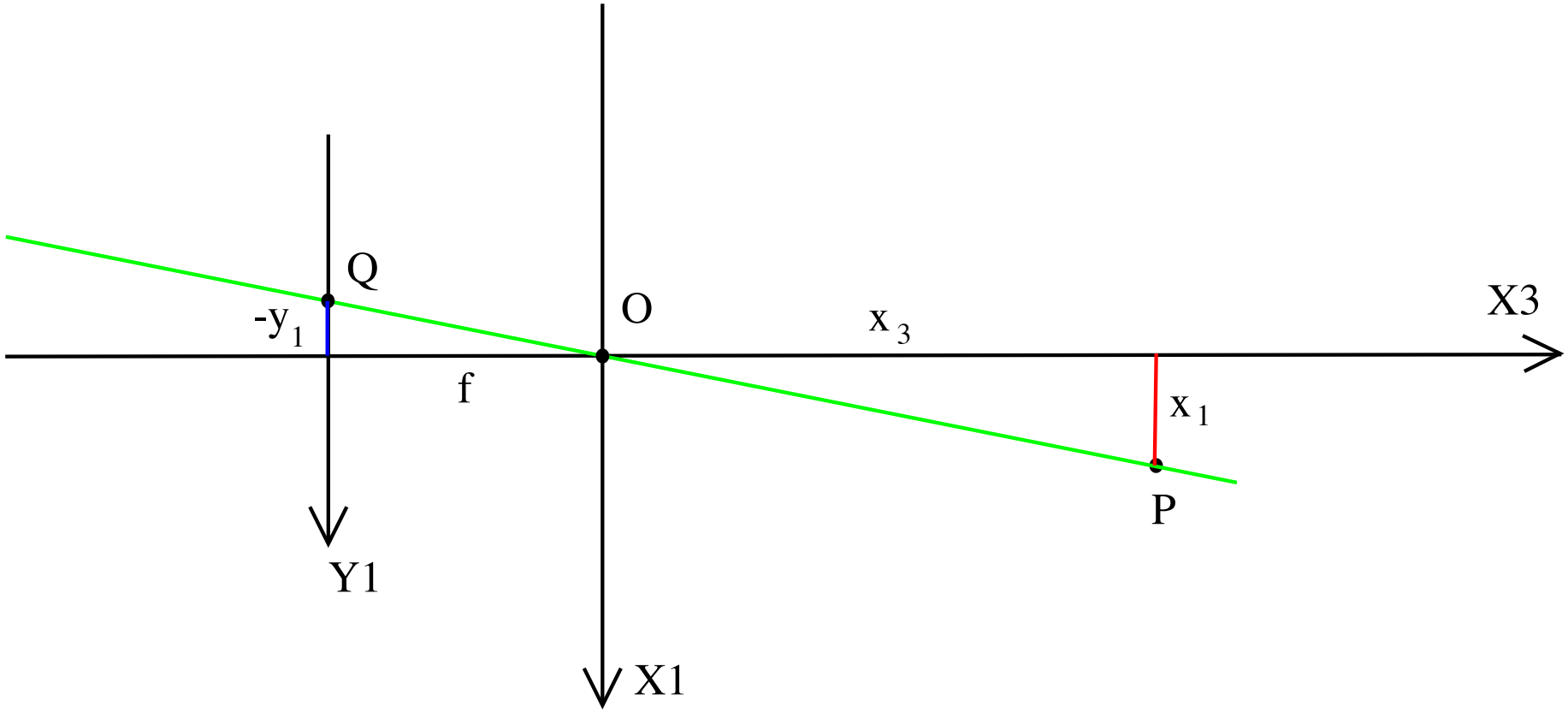


Image source: en:User:KYN, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pinhole2.svg

Pinhole camera

Can we undo the mirroring over the image plane axes?

Yes! Simply by flipping the sign along both axes:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \frac{f}{x_3} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

In a physical device, this would correspond to rotating the sensor by 180 degrees.



Pinhole camera

Can this projection be expressed in an augmented matrix (generally called *homogeneous coordinates*)?

Image source: https://www.pexels.com/photo/question-mark-on-paper-crafts-5428828/

Homogeneous coordinates

From Wikipedia, page "Homogeneous coordinates":

> In mathematics, **homogeneous coordinates** or **projective coordinates**, introduced by August Ferdinand Möbius in his 1827 work Der *barycentrische Calcul*, are a system of coordinates used in projective geometry, just as Cartesian coordinates are used in Euclidean geometry. They have the advantage that the coordinates of points, including points at infinity, can be represented using finite coordinates. Formulas involving homogeneous coordinates are often simpler and more symmetric than their Cartesian counterparts.

Homogeneous coordinates

From Wikipedia, page "Homogeneous coordinates":

> Given a point *(x,y)* on the Euclidean plane, for any non-zero real number *Z*, the triple *(xZ,yZ,Z)* is called a set of homogeneous coordinates for the point. By this definition, multiplying the three homogeneous coordinates by a common, non-zero factor gives a new set of homogeneous coordinates for the same point.

Homogeneous coordinates

We can recover the "true" coordinates by dividing the homogeneous coordinates of the point by the last (augmented) component, retrieving the coordinates in Euclidean plane.

The augmented coordinates in the Euclidean coordinate system should always have a 1 as the augmented component. Since the homogeneous coordinates have a common, non-zero scaling factor, division by this factor recovers the Euclidean coordinates.

Camera matrix

Start with the projection

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \frac{f}{x_3} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

and add the additional 1 coefficient for the augmented matrix:

$$\begin{bmatrix} y_1 \\ y_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{f}{x_3} x_1 \\ \frac{f}{x_3} x_2 \\ 1 \end{bmatrix}$$

Camera matrix

Divide by the focal length f:

$$\begin{bmatrix} y_1 \\ y_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{f}{x_3} x_1 \\ \frac{f}{x_3} x_2 \\ 1 \end{bmatrix} \sim \begin{bmatrix} \frac{1}{x_3} x_1 \\ \frac{1}{x_3} x_2 \\ \frac{1}{f} \end{bmatrix}$$

Note: $\sim$ means the matrices are equal given a non-zero scaling factor.

Camera matrix

Multiply by x_3:

$$\begin{bmatrix} y_1 \\ y_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{f}{x_3} x_1 \\ \frac{f}{x_3} x_2 \\ 1 \end{bmatrix} \sim \begin{bmatrix} \frac{1}{x_3} x_1 \\ \frac{1}{x_3} x_2 \\ \frac{1}{f} \end{bmatrix} \sim \begin{bmatrix} x_1 \\ x_2 \\ \frac{x_3}{f} \end{bmatrix}$$

Note: $\sim$ means the matrices are equal given a non-zero scaling factor.

Camera matrix

$$\begin{bmatrix} y_1 \\ y_2 \\ 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{f} & 0 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 1 \end{bmatrix}$$

or

$\mathbf{y} \sim C \cdot \mathbf{x}$ with C being called the *camera matrix*.

Note: $\sim$ means the matrices are equal given a non-zero scaling factor.

Camera matrix

The canonical camera matrix of the perspective projection is:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{f} & 0 \end{bmatrix} \sim \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Note: $\sim$ means the matrices are equal given a non-zero scaling factor.

Pinhole camera

We have completed our pinhole camera model:

$$\begin{bmatrix} y_1 \\ y_2 \\ 1 \end{bmatrix} \sim \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 1 \end{bmatrix}$$

What is the scaling factor of the homogeneous coordinates here?

Can the pinhole camera be done?



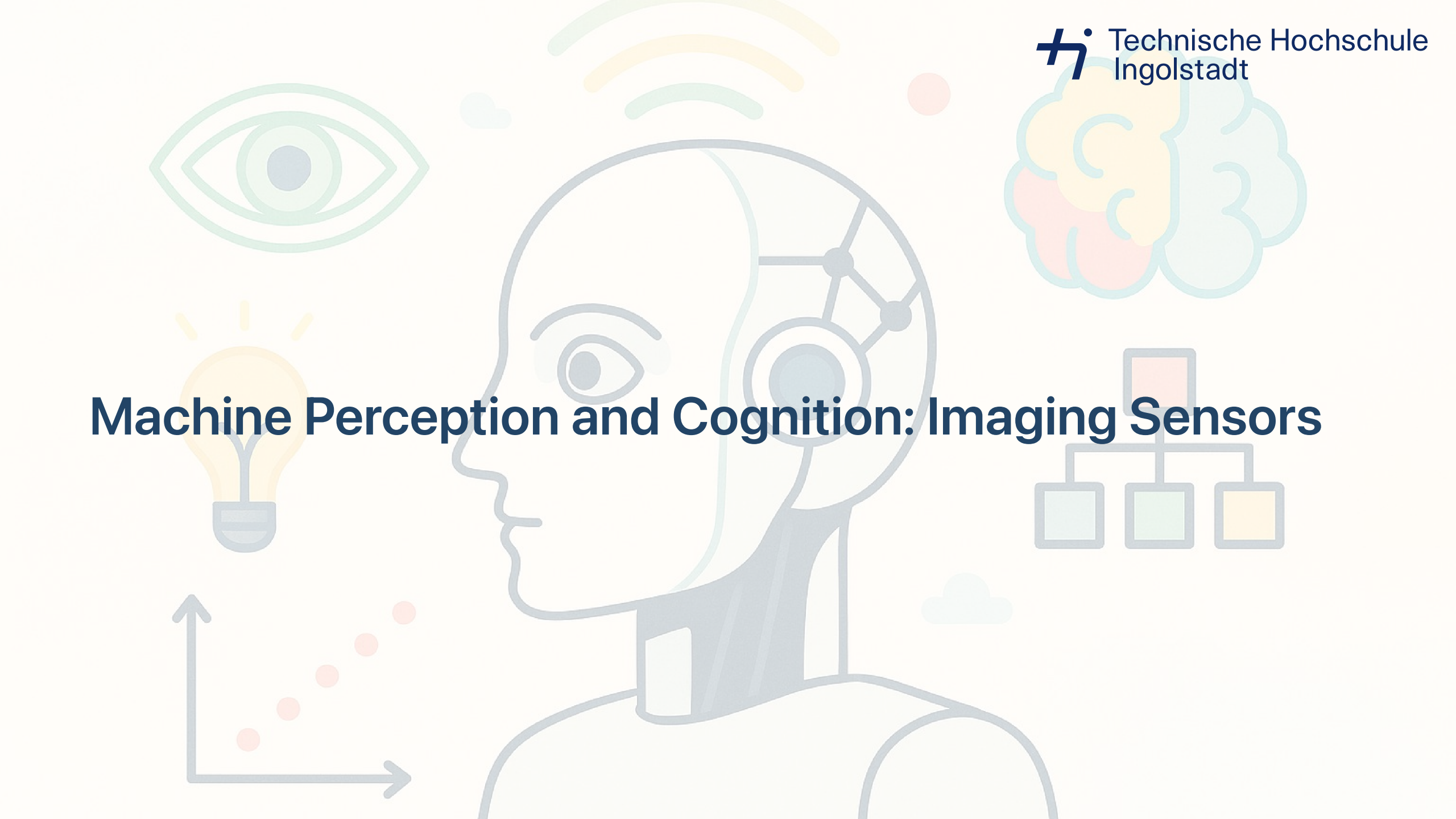
Image source: Ulli Purwin, CC BY 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:AV1-Lochkamera.JPG

Pinhole camera

Recap!

- 📍 Projections map 3D world points to 2D image points along projection lines.
- 👁️ Perspective projection emulates how the human eye sees the world.
- 🔭 Perspective projection shows objects that are farther away as smaller.
- 📸 Pinhole camera is a geometric model to generate a perspective projection.

Technische Hochschule Ingolstadt



Machine Perception and Cognition: Imaging Sensors



Imaging sensors

So how does the image get into the computer?

Image source: https://www.pexels.com/photo/question-mark-on-chalk-board-356079/

Camera obscura

Camera obscura is a natural or low-tech implementation of a pinhole camera. We need it digital!



Image source: Gampe, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Camera_obscura_Prague.jpg

Digital single-lens reflex camera

Closer to what we need!

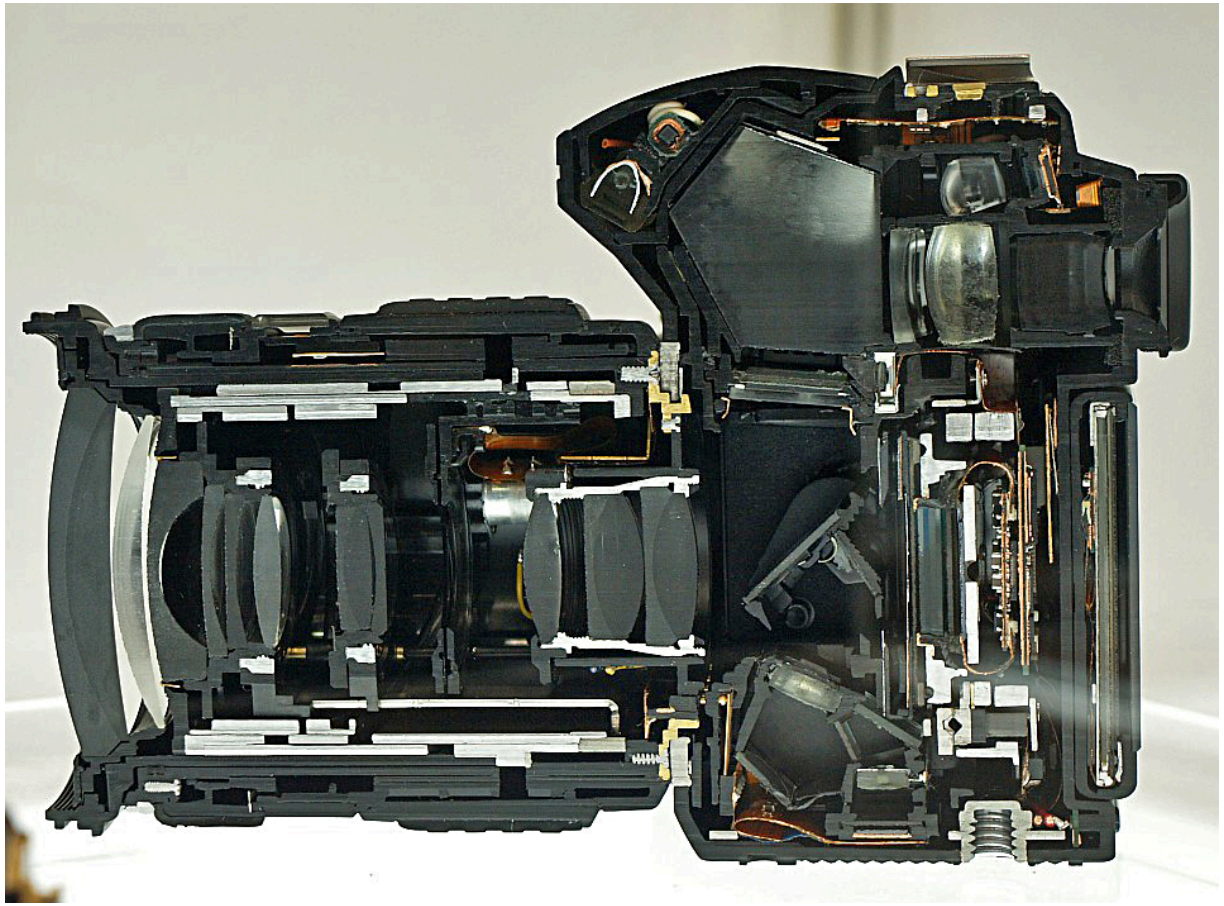


Image source: Hanabi123, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Olympus_E-30-Cutmodel.jpg

Rasterized image

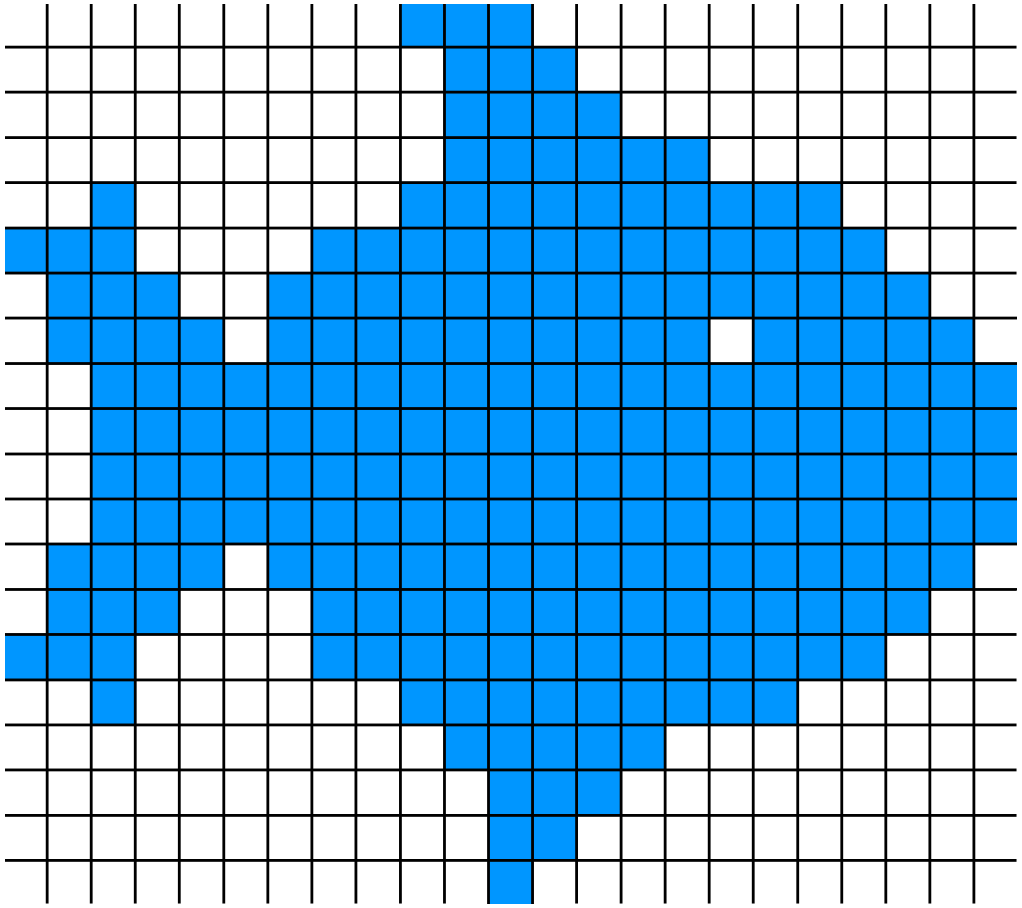


Image source: Andreas -horn- Hornig, de:Benutzer:Sjr, CC BY-SA 2.5 (https://creativecommons.org/licenses/by/2.5), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Raster_graphic_fish_20x23squares_sdtv-example.png

Rasterized image

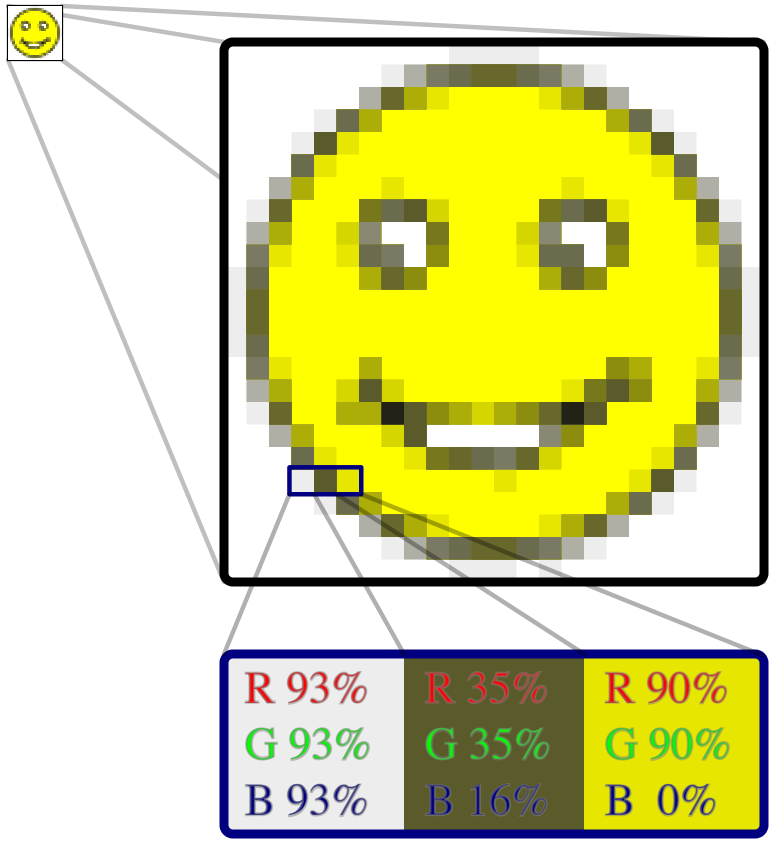


Image source: Gringer, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Rgb-raster-image.svg



What do we need?

We need a sensor that produces a rasterized image in such a way that more incoming light will produce higher values and less incoming light lower values.

Image source: https://www.pexels.com/photo/light-bulb-577514/

Imaging sensors

An imaging sensor is a device that forms the 2D image plane, collects light within a specific exposure time and senses the light intensity. Physically, each pixel (picture element) is a sensor on its own that integrates the incoming light over exposure time.

In digital cameras, there are two types of sensors:

- Charge-coupled device (CCD)
- Complementary metal-oxide-semiconductor (CMOS)

Photoelectric effect

Basis of the photodiode: Photons get absorbed and electrons emitted.

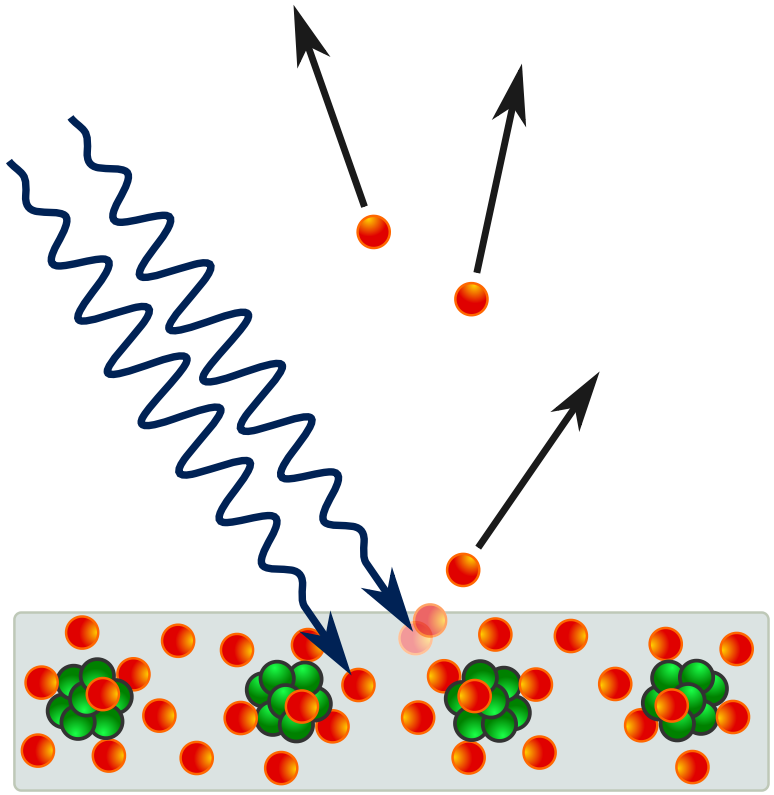


Image source: Ponor, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Photoelectric_effect_in_a_solid_-_diagram.svg

CCD sensors

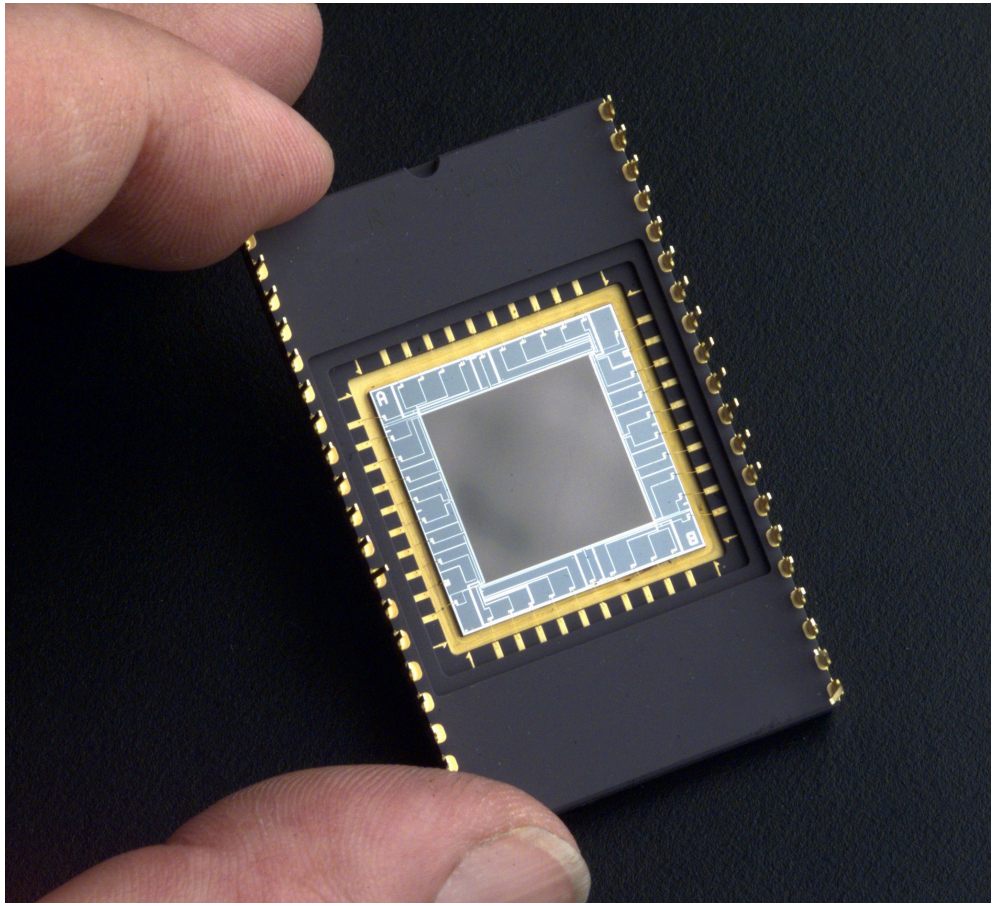


Image source: NASA, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Delta-Doped_Charged_Coupled_Devices_(CCD)_for_Ultra-Violet_and_Visible_Detection.jpg

CCD sensors

How does a CCD work?

Using the photoelectric effect, photons are collected and electrons emitted. Electrons are collected in a capacitor. The electrical charge is then moved from one pixel to the next (thus "coupled") during readout. Analog-digital conversion (quantization) is applied for quantization.

CCD sensors

How does a CCD work?

A CCD is a pixel-grid sensor where each pixel individually senses light:

1. Photons enter the pixel.
2. Electrons get emitted (photoelectric effect).
3. Electrons are collected in a capacitor.

CCD sensors

How does a CCD work?

Readout of the image proceeds as follows:

1. There is one amplifier per sensor, row or column.
2. The charged capacitor closest to the amplifier gets flushed and amplified.
3. Applying an analog-digital conversion produces a digital value.
4. The charge of the next closest capacitor gets moved.
5. Read the next value and repeat until all pixels are read.

CCD sensor

Working of a CCD (animated):

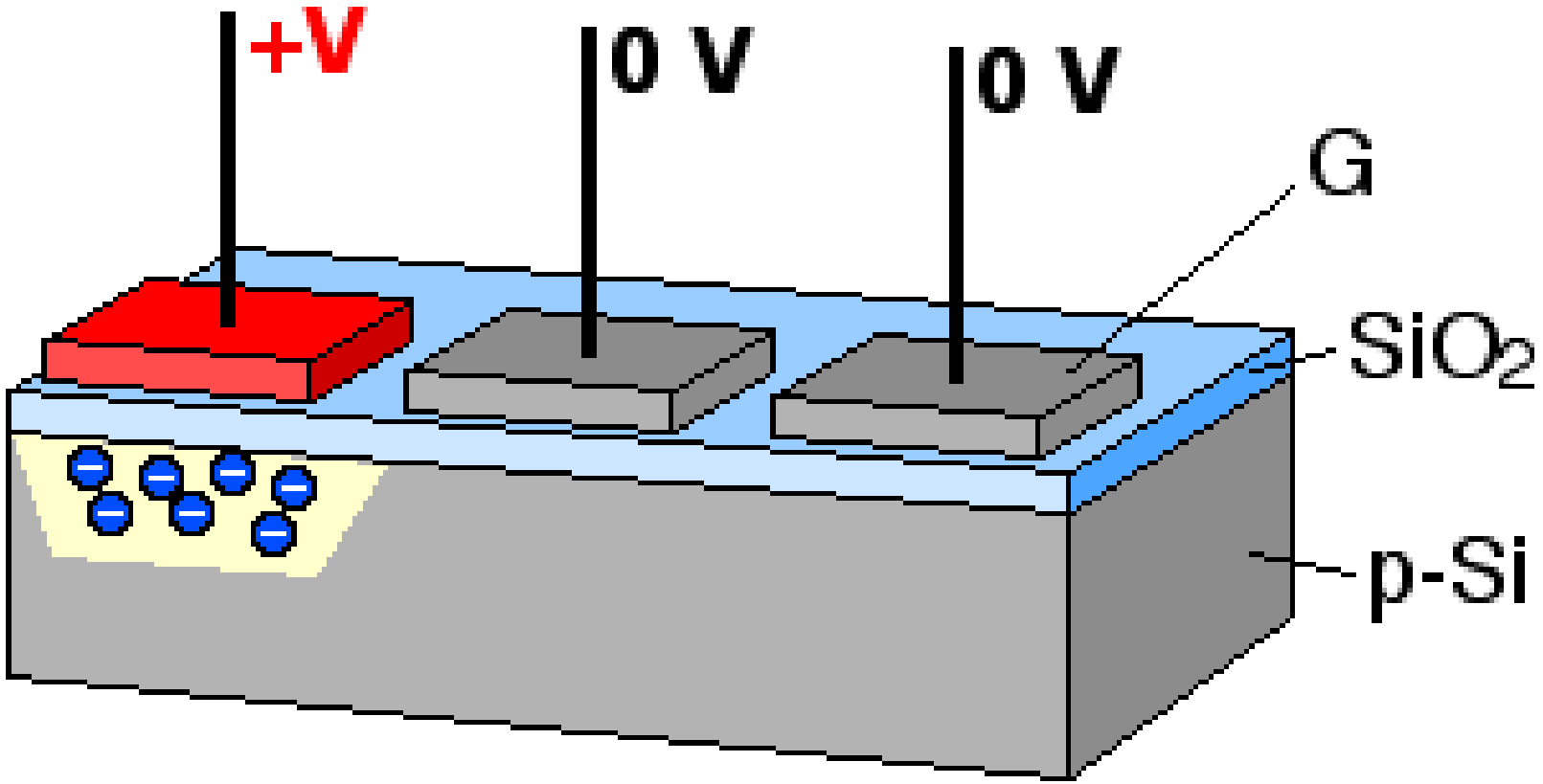


Image source: Michael Schmid, CC BY 2.5 (https://creativecommons.org/licenses/by/2.5), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:CCD_charge_transfer_animation.gif

CCD sensor

The capacitors of the pixels can "overflow" to neighboring pixels, producing blooming:

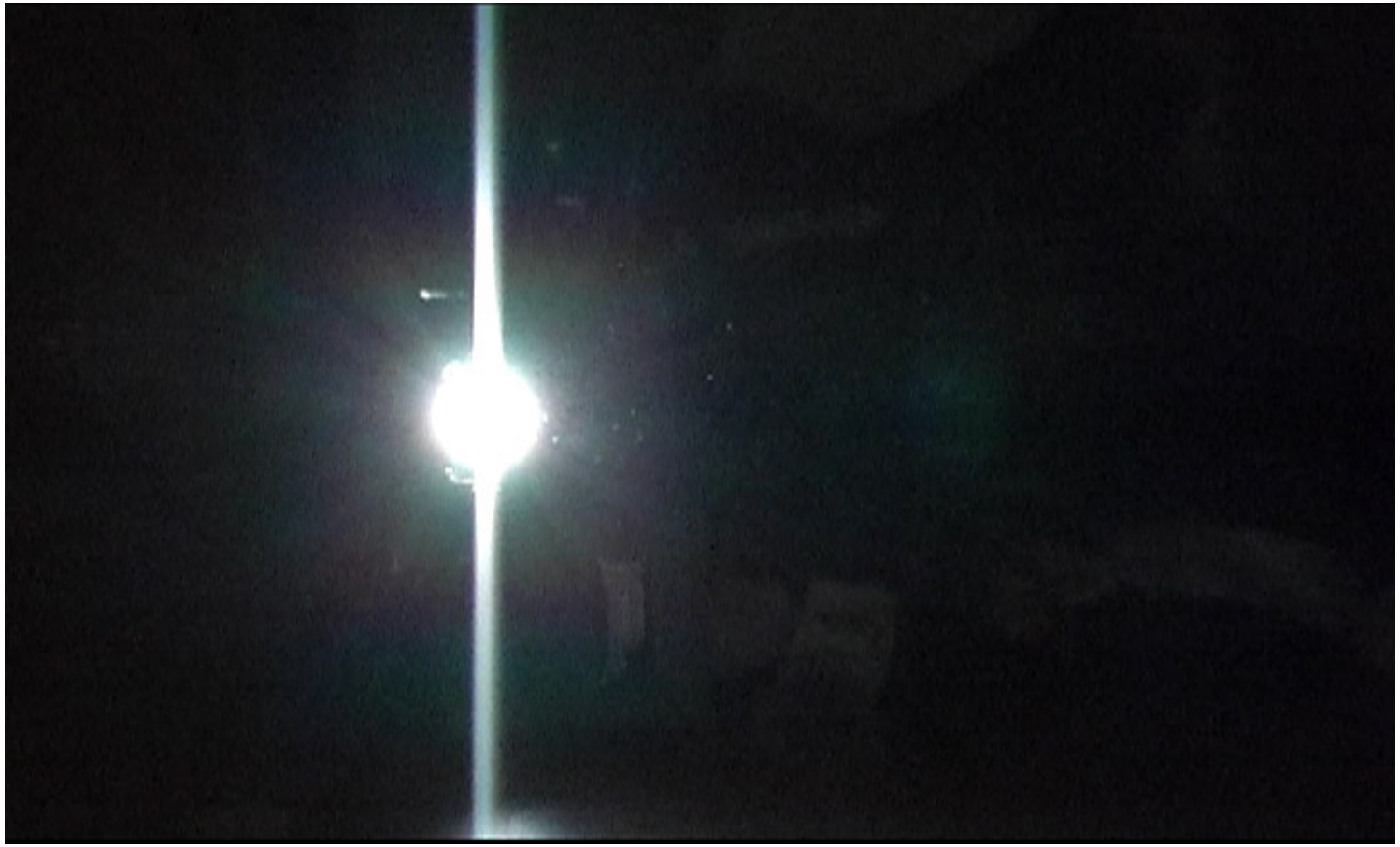


Image source: User:Genetichazzard, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Vertical_smear.jpg

CMOS sensors

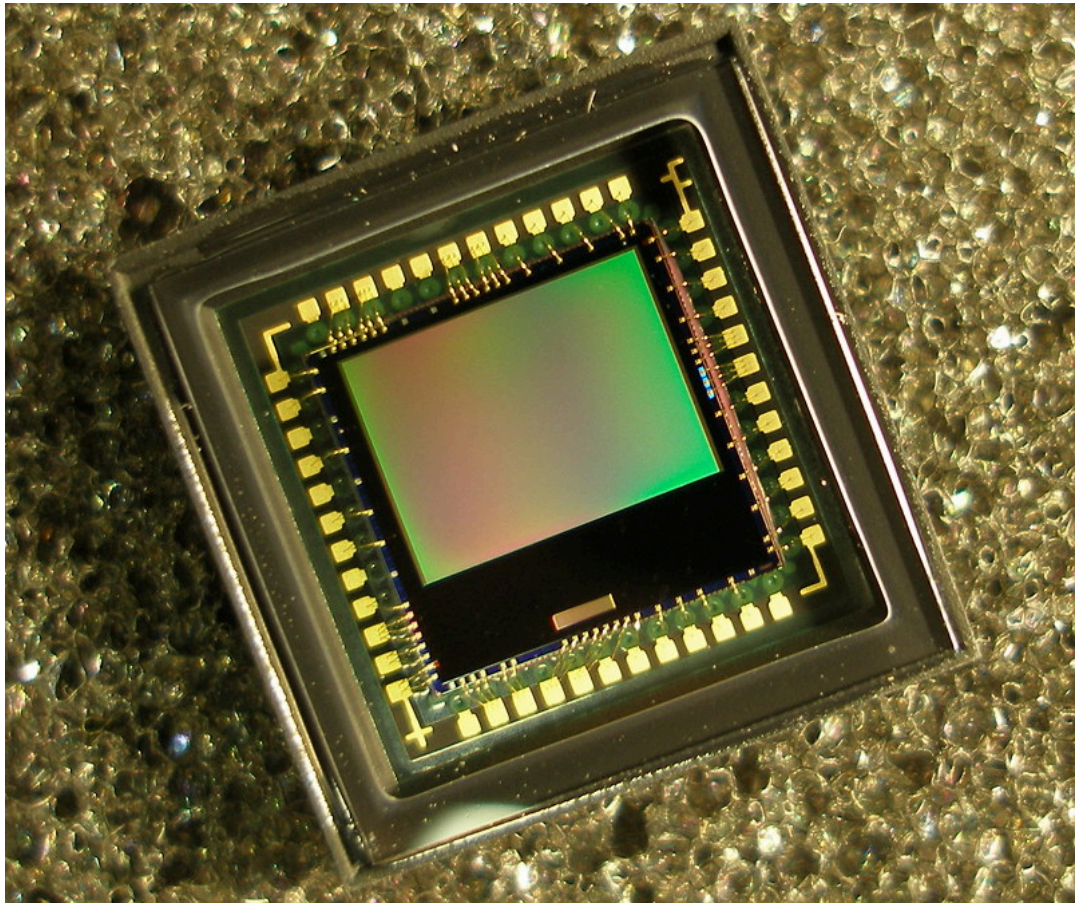


Image source: Filya1, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Matrixw.jpg

CMOS sensors

How does a CMOS work?

Each pixel has several transistors for control and analog-digital conversion or connecting it to a shared analog-digital converter. Using these, pixels are reset before exposure. Using the photoelectric effect, photons are collected and electrons emitted. Rows are then selected one-by-one and read by applying analog-digital conversion (quantization).

CMOS sensors

How does a CMOS work?

A CMOS is a pixel-grid sensor where each pixel individually senses light:

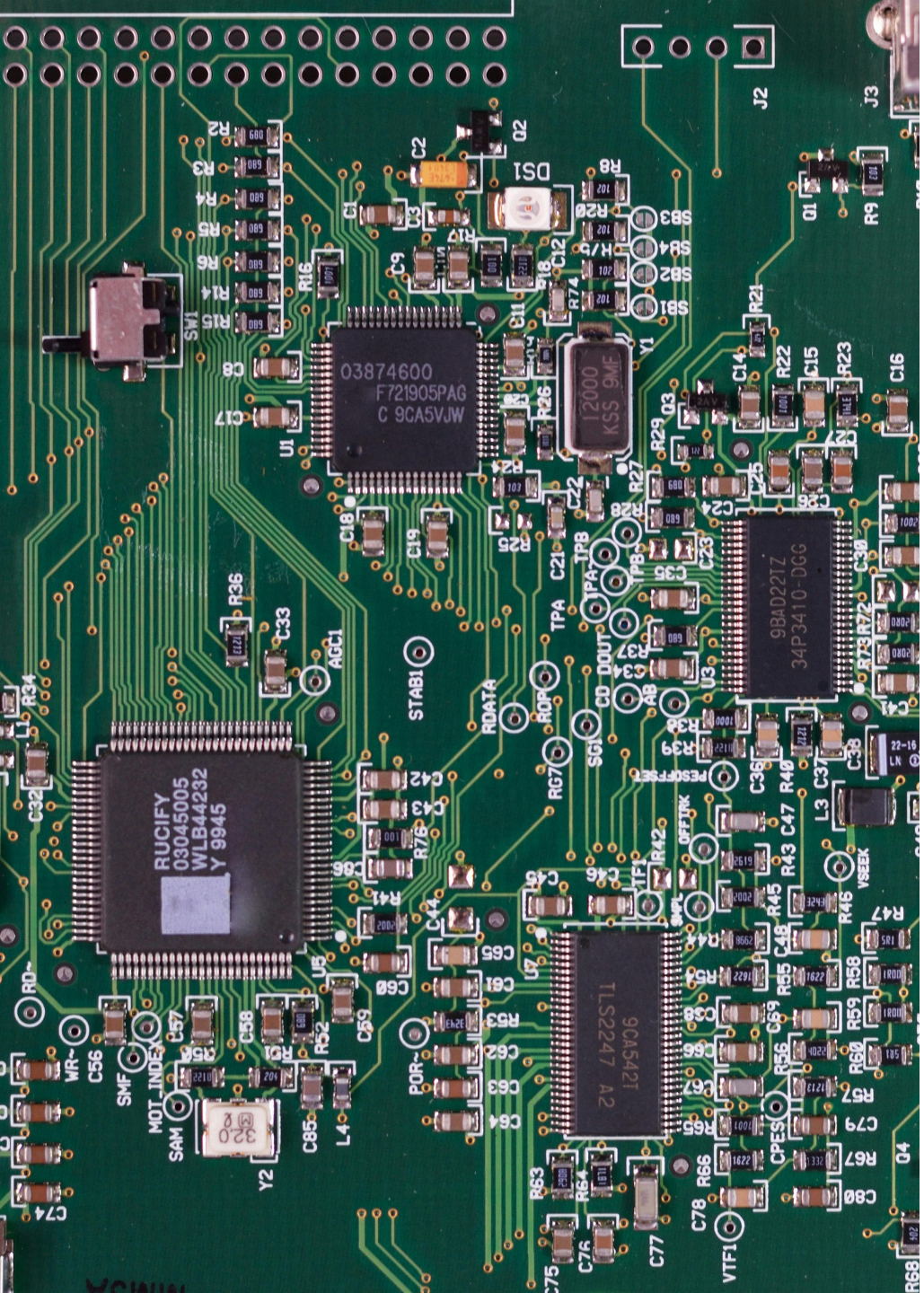
1. A transistor resets the pixel. This happens row-by-row!
2. Photons enter the pixel.
3. Electrons get emitted (photoelectric effect).
4. Electrons are collected in a capacitor.

CMOS sensors

How does a CMOS work?

Readout of the image proceeds as follows:

1. Each pixel has its own (partial) amplifier and is connected to a shared analog-digital converter.
2. Readout is done row-by-row.
3. The top-most row is selected and read.
4. Continuing with the next row and so on.



CMOS sensors

Some CMOS have additional transistors per pixel to "spill" excess charge and reduce blooming.

Image source: https://www.pexels.com/photo/green-circuit-board-343457/

CCD vs. CMOS

Pro CCD:

- Lower noise.
- Global shutter easily possible.

Pro CMOS:

- Cheaper to manufacture.
- Less prone to blooming.